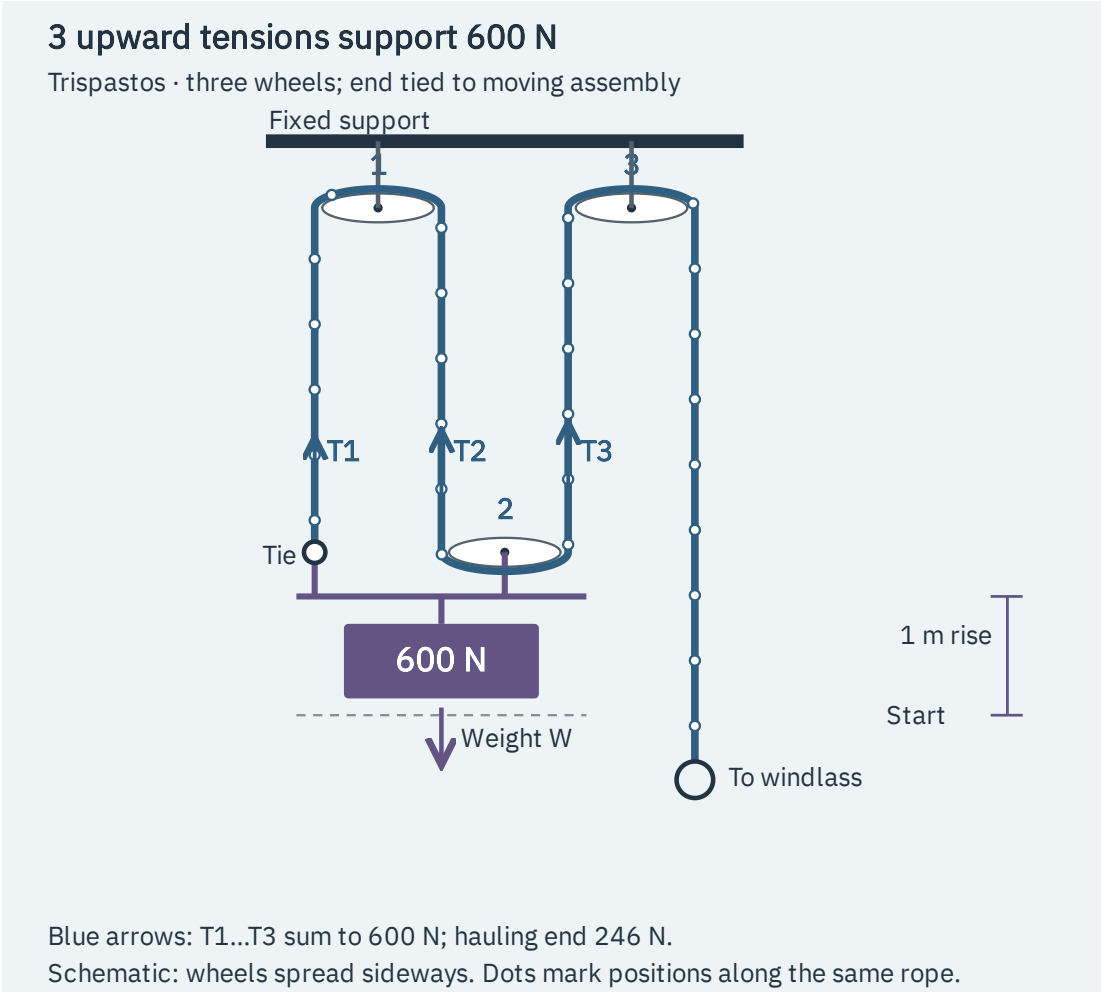
Where you tie the rope changes the lift.

Trispastos · three wheels; end on moving assembly; held load rise 1 m; weight 600 N; arm/drum ratio 2; chosen turning-wheel tension transfer 90%.

Edition 2fd3412c88917eb226cb871217e6d7da3fdf4292902172e1d063db00c7939154. Reopen these settings. This file embeds its fonts, diagrams, measurements, language notes and five full printed pages. It contains no scripts.

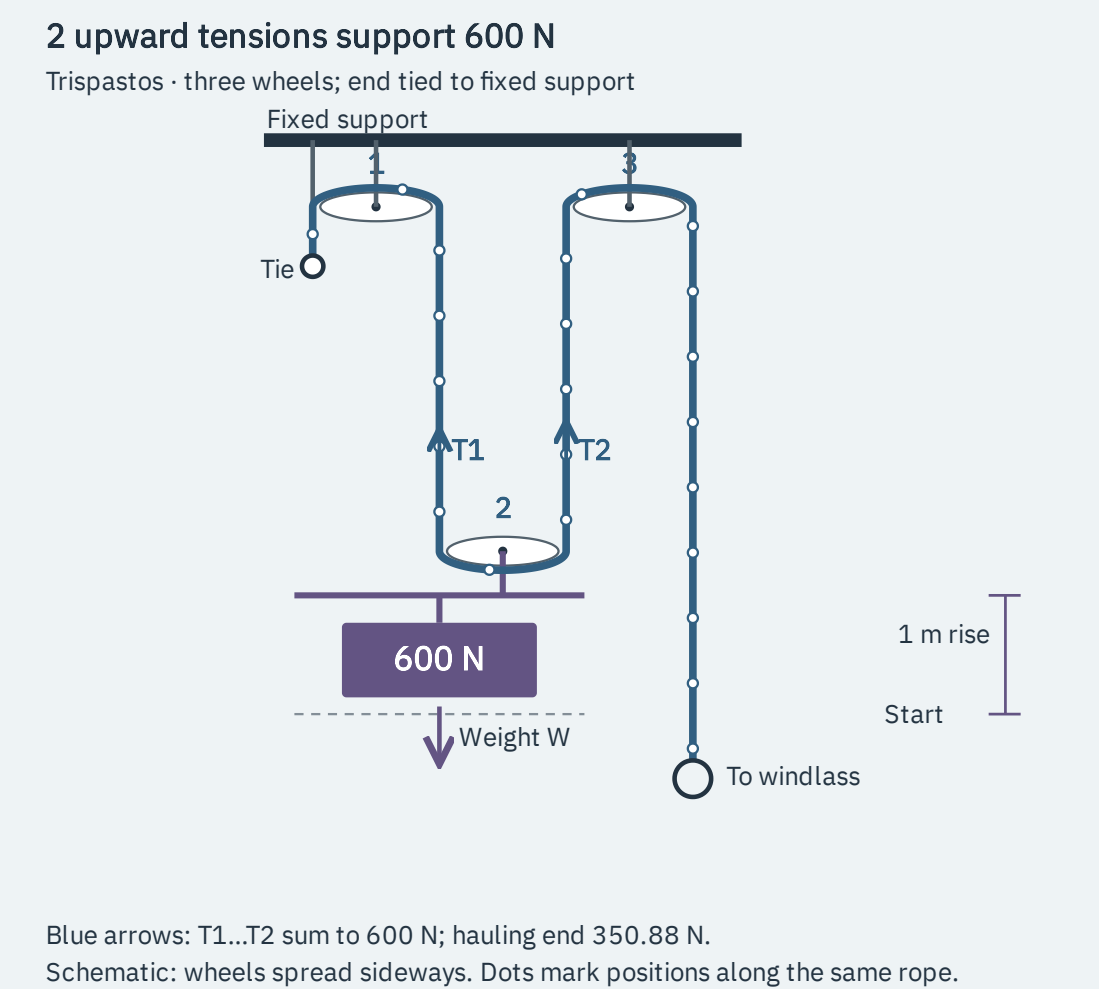
Follow the actual rope route



3 supporting portions: 179.335793 + 199.261993 + 221.402214 N upward. Hauling-end tension: 246.00246 N. The load rises 1 m while 3 m wind onto the drum. The total weight includes moving hardware.

The source ties the end to the lower block. A tie on the fixed support is our modern alteration. A single fixed wheel is a modern control. The wheels are spread sideways to expose the route; the source locates wheels within each block. The drawing is not a scale restoration of the frame.

Change only the tied end



The same wheel count now gives 2 supporting portions. The same numeric load rise is held fixed. All other settings stay the same.

The same rope. Unequal pulls.

Chosen tension transfer at each turning wheel: 90%. Whole-system efficiency: 81.3%. These are modern raising assumptions, not measurements of ancient machinery. The windlass stays ideal. The first upper wheel on the fixed-tie route does not turn, is assigned zero resisting torque and receives no modeled loss.

Where your work goes



Lifting (blue)

600 J

Loss (purple)

138.007 J

Input work

738.007 J

Whole system

81.3%

The whole bar represents your input work.

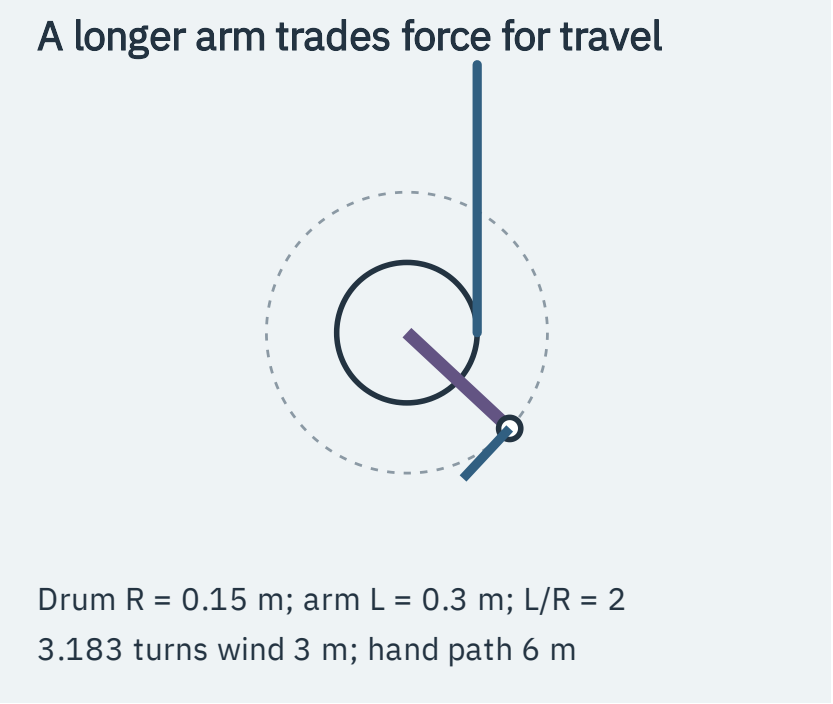
Wheels numbered from tied end to windlass. Tensions describe steady raising, including at zero selected travel.

Wheel ledger, raising

Wheel	Tied-end side (N)	Hauling side (N)	Rope travel relative to axle (m)	Dissipated work (J)
1, upper	179.3358	199.262	1	19.9262
2, lower	199.262	221.4022	2	44.2804
3, upper	221.4022	246.0025	3	73.8007

At each turning wheel, tied-end-side tension equals η times hauling-side tension. Loss is their tension difference times rope travel relative to the axle. Sum the wheel losses: 138.00738 J. Input work 738.00738 J = load work 600 J + dissipated work 138.00738 J. At zero travel all work is zero; the bar shows the proportions for a positive lift. Lowering, starting friction and actual ancient efficiencies are outside this experiment.

Turn the windlass



Chosen drum R = 0.15 m; arm L = 0.3 m; tangential hand force 123.00123 N. The hand path is 6 m and the drum makes 3.183099 turns.

Load: 600 N × 1 m ≈ 600 J.

Rope: 246.00246 N × 3 m ≈ 738.00738 J.

Hand: 123.00123 N × 6 m ≈ 738.00738 J.

With n supporting portions, each moving straight run shortens by the rise h; circular arcs retain their length. Thus rope take-up s = n h. With ideal wheels, equal tension T balances W: n T = W. With losses, the individually calculated upward tensions sum to W. The ideal windlass balances moments: F L = T R. Its hand path is (L/R)s, so T s = F × hand path. Only for ideal wheels does this also equal W h; otherwise input work includes the calculated wheel losses. Displayed values are rounded; the embedded CSV retains full floating-point values.

Compare the same controlled distance

The same weight rises through the same distance in every row. More supports require more rope; load work stays the same. Hauling tension and dissipated work depend on the chosen wheel losses.

Same load rise: 1 m; same weight 600 N and arm ratio 2. Chosen moving-wheel tension transfer 90%. Raising only.

Arrangement	Supports	Rise (m)	Rope (m)	Hauling tension (N)	Hand (N)	Hand path (m)	Load work (J)	Input work (J)	Loss (J)
One fixed wheel End on load; modern control	1	1	1	666.6667	333.3333	2	600	666.6667	66.6667
Trispastos · three wheels End on moving block; source route	3	1	3	246.0025	123.0012	6	600	738.0074	138.0074
Trispastos · three wheels End on fixed support; your alteration	2	1	2	350.8772	175.4386	4	600	701.7544	101.7544
Pentaspaston · five wheels End on moving block; source route	5	1	5	162.7962	81.3981	10	600	813.9809	213.9809
Pentaspaston · five wheels End on fixed support; your alteration	4	1	4	193.8548	96.9274	8	600	775.4192	175.4192

Read the Latin and quoted Greek

Two wheels in the upper block

X.2.1, pp.226–227

in trocleam induntur orbiculi ⟨II⟩ per axiculos versationes habentes.

Into the pulley block are set ⟨two⟩ wheels, able to turn on their small axles.

Krohn supplies II in angle brackets; the apparatus attributes the addition to Joc. The sentence crosses the printed page boundary. Read the next page as well. This is an edited witness, not a surviving ancient blueprint.

in trocleam: into the pulley block. In with an accusative indicates placement into it. Troclea is the containing block here; orbiculi are its rotating wheels.

orbiculi: small wheels / sheaves. Nominative plural, the subject of passive induntur. A wheel count alone does not state how many rope portions support a moving load.

per axiculos: on / by means of small axles. Per with the accusative describes the means of rotation. Axiculus is a diminutive of axis.

versationes habentes: having rotations. Habentes agrees with orbiculi; the wheels are able to turn.

Follow the rope to its tied end

X.2.1, p.227

per orbiculum ⟨summum⟩ traicitur ductarius funis, deinde demittitur et traducitur circa orbiculum trocleae inferioris, refertur autem ad orbiculum imum trocleae superioris et ita descendit ad inferiorem et in foramine eius religatur.

The hauling rope is passed over the ⟨uppermost⟩ wheel; then it is let down and led around the wheel of the lower block. It is brought back to the lowest wheel of the upper block, then descends to the lower block and is tied in its opening.

Summum is an editorial addition attributed to Joc. The decisive attachment is religatur: the end is tied to the lower block. The diagram spreads the upper wheels sideways to make the route legible; it does not reproduce their described vertical arrangement.

ductarius funis: hauling rope. Funis is nominative singular, the subject of the passive verbs that follow the rope's route. Ductarius describes its hauling use.

trocleae inferioris: of the lower block. Two genitives modify orbiculum. Inferioris contrasts with superioris, the upper block.

imum: lowest. Accusative masculine singular agrees with orbiculum. The text locates wheels within the upper block; our unrolled diagram separates them horizontally.

refertur: is brought back. Present passive of refero. Keep following the same rope, rather than imagining another independently pulled line.

in foramine eius: in its opening. In with the ablative locates the tie. Eius refers to the lower block just mentioned.

religatur: is tied / fastened. Present passive of religo. This attachment moves with the lower assembly and supplies one of its upward tensions.

Wind the hauling end

X.2.1–2, p.227

altera pars funis refertur inter imas machinae partes. … cum autem funis habet caput ad suculam religatum et vectes ducentes eam versant, funis ⟨se⟩ involvendo circum suculam extenditur et ita sublevat onera ad altitudinem et operum conlocationes.

The other part of the rope is brought between the machine's lowest parts. ... When the rope has its end fastened to the windlass, and the bars that drive it turn it, the rope, by winding (itself) around the windlass, is pulled taut and so raises loads to the height and positions required by the works.

The ellipsis omits the intervening description of the windlass mounting and stone grips. Krohn adds se. The circular detail shows a chosen constant-radius drum and perpendicular hand force; these dimensions and the numerical force law are modern analysis.

altera pars: the other part. Altera distinguishes the hauling side from the end already fastened to the moving block.

ad suculam religatum: fastened to the windlass. Religatum agrees with caput, the rope's end. Do not confuse this second fastening with the tie to the lower block.

vectes: bars / levers. Nominative plural, the subject of versant. Eam refers to the windlass they turn.

involvendo: by winding. Ablative gerund gives the means. Circum with the accusative suculam says what the rope winds around.

onera: loads. Accusative plural of onus, the object of sublevat. The model's weight is a force in newtons, not a Roman unit of mass.

Three wheels; five wheels

X.2.3, p.227

haec autem ratio machinationis, quod per tres orbiculos circumvolvitur, trispastos appellatur. cum vero in ima troclea duo orbiculi, in superiore tres versantur, id pentaspaston dicitur.

This arrangement of machinery, because the rope winds around three wheels, is called trispastos. When two wheels turn in the lower block and three in the upper, it is called pentaspaston.

These names count wheels in the described arrangements. Our fixed-support tie is a modern alteration for comparison, not a second ancient prescription. A single fixed wheel is a modern control example.

per tres orbiculos: around / through three wheels. Per with the accusative describes the rope's route. The following clause specifies how the five wheels are divided between the two blocks.

in ima troclea: in the lower block. In with the ablative locates two wheels in the lower block. Three are in superiore, the upper one, with troclea understood.

trispastos: the named three-wheel arrangement. A Greek technical name in the Latin text. The experiment retains Krohn's Latin spelling rather than presenting a back-translated Greek quotation.

pentaspaston: the named five-wheel arrangement. Another Greek technical name as printed in Latin. In the source route the tied end adds a fifth supporting tension to the four provided by two moving wheels.

Straight and circular motion

X.3.1, p.232

una porrecti, quam Graeci εὐθεῖαν vocitant, altera rotunditatis, quam Graeci κυκλωτήν appellant.

One is straight extension, which the Greeks call eutheian; the other is circular motion, which the Greeks call kyklōtēn.

This excerpt selects the two kinds of motion from the preceding sentence. Krohn prints the Greek forms; his apparatus attributes κυκλωτήν to Joc. and records Latin-letter manuscript variants. These are forms in this edition, not an independent Greek treatise.

porrecti: of straight extension. A genitive form used for the straight kind of motion. This is contrasted with rotunditatis, circularity.

εὐθεῖαν: straight (form in the Latin text). Accusative feminine singular of εὐθύς, eutheian in transliteration. A noun is understood in the Greek expression. The form is quoted, not a label for an ancient numerical formula.

κυκλωτήν: circular (edited form). Accusative feminine singular, kyklōtēn. Krohn's apparatus credits this Greek reading to Joc.; the Latin-letter witnesses differ. Compare the printed apparatus before treating the spelling as unproblematic.

Rotation produces a straight lift

X.3.2, p.232

inducuntur uti centra axiculi in orbiculos et in trocleis conlocantur, per quos orbiculos funis circumactus directis ductionibus et in sucula conlocatus vectium versationibus onerum facit egressus in altum.

Small axles are inserted as centres into the wheels and placed in the pulley blocks. The rope, passed around these wheels with straight runs and set on the windlass, brings loads upward through the turning of the bars.

The source links turning and straight hauling. In the model, rope wound onto the drum equals the shortening of the drawn rope route. Stored rope is reported as a length; the illustration does not model additional layers changing the drum radius.

uti centra: as centres. Uti means 'as' here. The small axles occupy the wheels' centres of rotation.

directis ductionibus: with straight pulls / runs. Ablative plural. It connects the rope's straight portions with the wheels' circular action.

vectium versationibus: by the turning of the bars. Vectium is genitive plural of vectis; versationibus is ablative plural, expressing the means.

A longer arm changes the effort

X.3.2–3, pp.232–233

quod Graeci ὑπομόχλιον appellant … id autem ⟨eo facilius fit⟩, quo brevior pars prior vectis ab ea pressione, quod est centrum, subit sub onus, et quo longius ab eo centro distans caput eius deducitur.

What the Greeks call hypomochlion … this ⟨is done more easily⟩ the shorter the front part of the lever that goes beneath the load from the support, which is the centre, and the farther its end is brought down from that centre.

The ellipsis joins the fulcrum name at the end of p.232 to the arm comparison on p.233; it does not imply adjacent words. Krohn supplies eo facilius fit. Applying the ideal lever law to a constant-radius windlass is our mathematical comparison, not a quoted ancient calculation.

ὑπομόχλιον: fulcrum / support. Hypomochlion, a neuter Greek noun quoted in the Latin sentence. On p.233 it appears again in Latin letters after supra. The windlass's axle is the centre in our circular comparison.

brevior: shorter. Comparative agrees with pars, the part of the lever on the load side.

quo … et quo: the more … and the more. The paired comparative clauses describe shortening the load arm and increasing the effort arm. They do not give the displayed modern metre values.

These are excerpts. Line-end hyphens are joined; angle brackets retain editorial additions; … marks omissions. The first sentence crosses pp.226–227. The final excerpt joins the fulcrum term on p.232 to the arm comparison on p.233. The five source pages preserve the complete surrounding text and apparatus.

Source and model boundaries

An ideal, quasi-static model with a massless inextensible rope, parallel vertical portions and either ideal wheels or a chosen constant tension-transfer ratio at each turning wheel during raising. The stated weight includes the load and moving hardware. No ancient efficiency is supplied by the text. A non-turning wheel is assigned zero resisting torque. The windlass remains ideal. Frame geometry, strength, starting friction, lowering, acceleration, rope layers on the drum and operator capacity are outside the model. The dimensions, newtons and metres are modern teaching choices, not measured ancient equipment.

Vitruvius, *De architectura*, X.2.1–3 and X.3.1–3, ed. F. Krohn, Leipzig: B. G. Teubner, 1912. Original public-domain volume. Krohn's upper-wheel count II and summum are additions attributed to Joc.; his

apparatus also distinguishes the Greek κυκλωτήν reading from Latin-letter witnesses. Nothing here repairs those textual layers silently.

The ideal numerical laws are modern analysis; OpenStax's simple-machines discussion explains the background. Historical text and scans: public domain. Code, new English, language notes and diagrams: PolyForm Noncommercial, Anthony Mercer, 2026. Every embedded scan, font and license was verified against its edition's SHA-256 identity before export.

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Krohn 1912, title page; PDF page 5. Title and 1912 imprint.

earum versationibus, nonnulla organis, et ita quae animadverterunt ad usum utilia esse studiis, artibus, institutis, gradatim augenda doctrinis curaverunt. attendamus enim primum inventum de necessitate, ut vestitus, quemadmodum telarum organicis administrationibus conexus staminis ad subtemen non modo corpora tegendo tueatur, sed etiam ornatus adiciat honestatem. cibi vero non habuissemus abundantiam, nisi iuga et aratra bubus iumentisque omnibus essent inventa. sucularumque et prelorum et vectium si non fuisset torculariis praeparatio, neque olei nitorem neque vitium fructum habere potuissemus ad iucunditatem, portationesque eorum non essent, nisi plostrorum seu serracorum per terram, navicularum per
6 aquam inventae essent machinationes. trutinarum vero librarumque ponderibus examinatio reperta vindicat ab iniquitate iustis moribus vitam. non minusque sunt innumerabili modo rationes machinationum, de quibus non necesse videtur disputare, quod non sunt ⟨ignotae sed⟩ ad manum cotidianae, ut sunt molae, folles fabrorum, raedae, cisia, torni ceteraque, quae communes ad usum consuetudinibus habent opportunitates.

Itaque incipiemus de îs, quae raro veniunt ad manus, 24
II ut nota sint, explicare. primumque instituemus de îs, quae aedibus sacris ad operumque publicorum perfectionem necessitate comparantur. quae fiunt ita. tigna duo ad onerum magnitudinem ratione expediuntur. a capite ea fibula coniuncta et in imo divaricata eriguntur, funibus in capitibus conlocatis et circa dispositis erecta retinentur. alligatur in summo troclea, quem etiam nonnulli rechamum dicunt. in trocleam induntur orbiculi ⟨II⟩ per axiculos

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1 itaq̄ *x* 6 tueatur *Schn*: -antur *x* 10 torcularis *x* 12 plaustrorum GS^c 15 exanimatio *G* umdicat *H* 16 minus quae (quę *GS*) *x* innumerabiles moderationes *G* 18 *add. Kr* 19 molae *Ro*: motae *x* 21 oportunitates GS^c: opportunitatib: *H* 23 sint *S*: sunt *HG* 26 magnitudinē *GS*: magnitudine HS^c ea fib. *Ro*: afib. *x* 28 alligatur . . . troclea *Joc*: alligantur . . . trocleae *x* 29 quem *ed*: quae *H*, quę SG^c, q̄ *G* 30 induntur *Joc*: induuntur *x* II (duo) *add. Joc*

Krohn 1912, printed p. 226; PDF page 244. X.2.1 begins; the upper wheels.

versationes habentes. per orbiculum ⟨summum⟩ traicitur ductarius funis, deinde demittitur et traducitur circa orbiculum trocleae inferioris. refertur autem ad orbiculum imum trocleae superioris et ita descendit ad inferiorem et in foramine eius religatur. altera pars funis refertur inter imas machinae partes. in quadris autem tignorum 2 posterioribus, quo loci sunt divaricata, figuntur chelonia, in quae coiciuntur sucularum capita, ut faciliter axes versentur. eae suculae proxime capita habent foramina bina ita temperata, ut vectes in ea convenire possint. ad rechamum autem imum ferrei forfices religantur, quorum dentes in saxa forata accommodantur. cum autem funis habet caput ad suculam religatum et vectes ducentes eam versant, funis ⟨se⟩ involvendo circum suculam extenditur et ita sublevat onera ad altitudinem et operum conlocationes. haec autem ratio machinationis, quod per tres orbiculos 3 circumvolvitur, trispastos appellatur. cum vero in ima troclea duo orbiculi, in superiore tres versantur, id pentaspaston dicitur. sin autem maioribus oneribus erunt machinae comparandae, amplioribus tignorum longitudinibus et crassitudinibus erit utendum ⟨et⟩ eadem ratione in summo fibulationibus, in imo sucularum versationibus expediendum. his explicatis antarii funes ante laxi conlocentur, retinacula super scapulas machinae longe disponantur, et si non erit, ubi religetur, pali resupinati defodiantur et circum fistucatione solidentur, quo funes alligentur. troclea in summo capite machinae rudenti contineatur, 4 et ex eo funis perducatur ad palum et quae est in palo trocleam inligata. circa eius orbiculum funis indatur et referatur ad eam trocleam, quae erit ad caput machinae

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1 summum *add. Joc* 2 dimittitur *G* 5 religatur *ed*: -gantur *x* 6 interim has *GS et a. c. H* 7 chelonia *ed*: helonia *x* 13 ad *HG*: ac *S* 14 *add. Joc* 18 pentas paston *HS*, pentes paston *G* 21 grassitudinibus *H* *add. Joc* 22 succularum *H* versationibus *Joc*: venationibus *x* 23 anterii *S* 28 funis perducatur *Rode*: funes pducantur *x* 29 indatur *HS*: inducatur *G*

Krohn 1912, printed p. 227; PDF page 245. Rope route, tied end, windlass and names.

que, a Paro, Proconnenso, Heraclea, Thaso uti marmor peteretur, propulsis ovibus Pixodarus in eodem loco pecus pascebat, ibique duo arietes inter se concurrentes alius alium praeterierunt et impetu facto unus cornibus percussit saxum, ex quo crusta candidissimo colore fuerat deiecta. ita Pixodarus dicitur oves in montibus reliquisse et crustam cursim Ephesum, cum maxime de ea re ageretur, detulisse. ita statim honores decreverunt ei et nomen mutaverunt: pro Pixodaro Euangelus nominaretur. hodieque quotmensibus magistratus in eum locum proficiscitur et ei sacrificium facit, et si non fecerit, poena tenetur.

III De tractoriis rationibus quae necessaria putavi, breviter exposui. quorum motus et virtutes duae res diversae et inter se dissimiles ⟨efficiunt⟩, ubi congruentes uti principia pariunt eos perfectus: una porrecti, quam Graeci ευθειαν vocitant, altera rotunditatis, quam Graeci κυκλωτην appellant. sed vero neque sine rotunditate motus porrecti nec sine porrecto rotationis versationes onerum possunt facere leva-
2 tiones. id autem ut intellegatur, exponam. inducuntur uti centra axiculi in orbiculos et in trocleis conlocantur, per quos orbiculos funis circumactus directis ductionibus et in sucula conlocatus vectium versationibus onerum facit egressus in altum. cuius suculae cardines uti centra porrecti in cheloniis, foraminibusque eius vectes conclusi capitibus ad circinum circumactis torni ratione versando faciunt oneris elationes. quemadmodum etiam ferreus vectis cum est admotus ad onus, quod manuum multitudo non potest movere, supposita uti centro ⟨ultro⟩ citro porrecta pressione, quod Graeci υπομοχλιον appellant, et lingua

1 proconesso *x* Thaso *ed*: thasii *HG*, tasu *S* marmor peteretur *Mar*: marmo|repã *H*, marmore petatur *GS* 4 petierunt G^c 6 reliquiss& crustam *H* 15 *add. Kr* ubi *Kr*: uti *x* 16 unā *x* eutheiā (-am) *x* 17 rotunditatis *S*: tutunditatis *H*, rotutunditatis *G* Graeci *om. S* κυκλωτήν *Joc*: cycletoen *HG*, cycleon *S* 25 cheloniis *S*: caeloniis *HG* 27 vectis *G*: -tes *HS* 29 ⟨ultro⟩ citro *Kr*: cito *x* 30 hypamoclion *HG*, -clyon *S*

sub onus subdita, caput eius unius hominis viribus pressum id onus extollit. id autem ⟨eo facilius fit⟩, quo brevior 3 pars prior vectis ab ea pressione, quod est centrum, subit (5) sub onus, et quo longius ab eo centro distans caput eius 5 deducitur. per id faciundo motus circinationis cogit pressionibus examinare paucis manibus oneris maximi pondus. item si sub onus vectis ferrei lingula subiecta fuerit neque eius caput pressione in imum, sed adversus in altitudinem extolletur, lingula fulta in areae solo habebit eam 10 pro onere, oneris autem ipsius angulum pro pressione. ita non tam faciliter quam per oppressionem, sed adversus nihilominus pondus oneris erit excitatum. igitur si plus lingula vectis supra hypomochlion posita sub onus subierit et caput eius propius centrum pressiones habuerit, non 15 poterit onus elevare, nisi, quemadmodum supra scriptum est, examinatio vectis longitudinis per caput neque ⟨in imum⟩ ductionibus fuerit facta.

Id autem ex trutinis, quae staterae dicuntur, licet con- 4 siderare. cum enim ansa propius caput, unde lancula 20 pendet, ibi ut centrum est conlocata et aequipondium in alteram partem scapi, per puncta vagando quo longius paulatim ad extremum perducitur, paulo et inpari pondere amplissimam pensionem parem perficit per scapi librationem, et examinato longius ab centro recedens ista 25 inbecillior aequipondii brevitas maiorem vim ponderis momento deducens sine vehementia molliter ab imo susum 254 versum egredi cogit [futurum]. quemadmodum etiam navis 5 onerariae maximae gubernator ansam gubernaculi tenens,

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1 subposita *S* 2 *add. Kr (Schn)* quo *Kr*: quod *x* 5 deducitur per id *Ro*: per id deducitur *x* 7 ferreis *S et a. c. G* 9 extolletur *rec*: -litur *x* 12 pondus *Kr*: inpondus (*cum* in *add.*) *x* excitatum *Joc*: exercitatum *x* 13 hypomocliū *x* 14 proprius *H* 16 *add. Kr* (*cf. v. 8*) 19 lancula *H*: lanuncula *S*, lingula *G* 20 ibi *Mar*: ubi *x* 22 paulatim *Kr*: aut etiam *x* et impari *Rode*: etiam pari *x* 23 parem *Joc*: parte *x* 24 examinato *Kr*: -tio *x* ista *Kr*: ita *x* 25 inbecilliora equipondii *H*, imbecilliora ęquipondii *GS* 27 *del. Joc* (*cf. 218, 4*) sursum *a. c. G* navis HG^c: naves H^cGS